LAPORAN

WORKSHOP DEVELOPER OPERATIONAL

MINGGU KE- 11

MATERI PEMBELAJARAN

Kubernetes Basic Setup & Learning Guide

Oleh

E41230153 Gilang Bayu Prasetyo

PROGRAM STUDI TEKNIK INFORMATIKA
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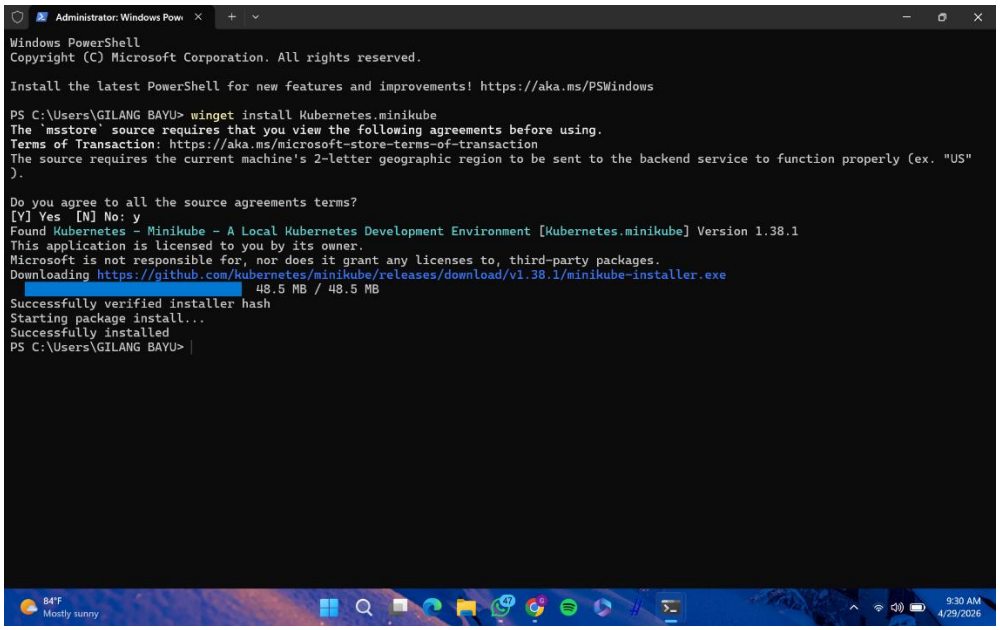
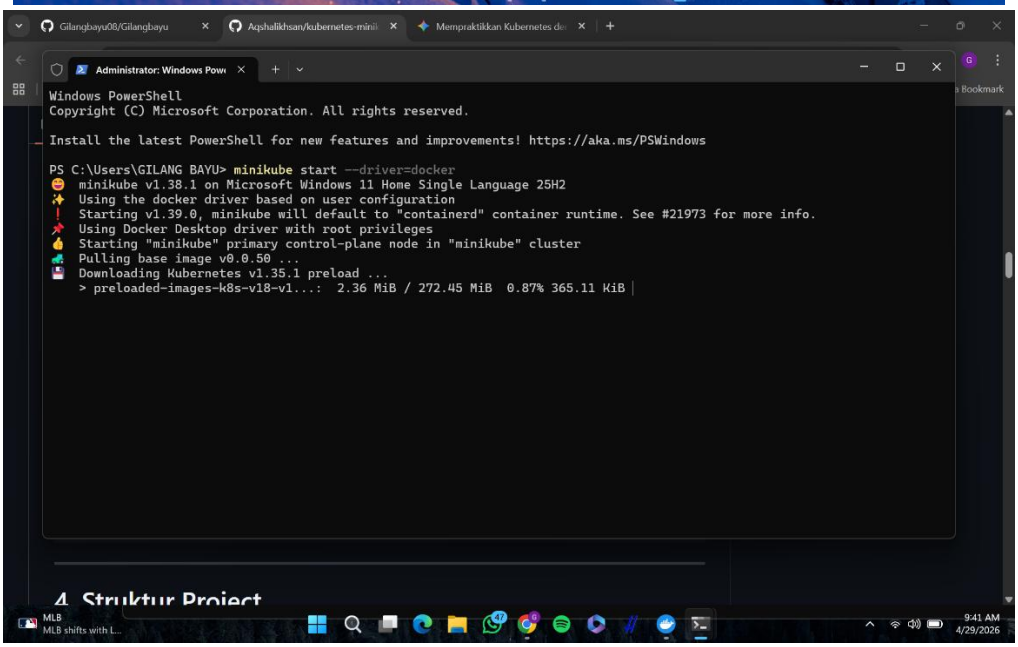
1. Instalasi & Inisialisasi Cluster

   Tahap awal adalah memastikan lingkungan Kubernetes (Minikube) siap digunakan di atas Docker Desktop.






```
gilang31@LAPTOP-F9B95IVJ:~$ kubectl version --client
Client Version: v1.34.1
Kustomize Version: v5.7.1
```

```
gilang31@LAPTOP-F9B95IVJ:~$ kubectl get nodes
NAME       STATUS   ROLES           AGE     VERSION
minikube   Ready    control-plane   6m53s   v1.35.1
gilang31@LAPTOP-F9B95IVJ:~$
```

2. Menyiapkan folder khusus di dalam WSL Ubuntu agar semua file konfigurasi tersimpan dengan rapi.

```
gilang31@LAPTOP-F9B95IVJ:~/projects/kubernetes-practice$ cd /mnt/e/kubernetes-minikube
```

deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: web-nginx
spec:
  replicas: 3
  selector:
    matchLabels:
      app: web-nginx
  template:
    metadata:
      labels:
        app: web-nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
```

```yaml
! service.yaml ×

! service.yaml
apiVersion: v1
kind: Service
metadata:
  name: web-nginx-service
spec:
  selector:
    app: web-nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
  type: NodePort
```

3. Deploy Aplikasi (Object Deployment)
   Mendaftarkan konfigurasi aplikasi ke dalam Kubernetes agar sistem menjalankan kontainer

```
gilang31@LAPTOP-F9B95IVJ:~/projects/kubernetes-practice$ cd /mnt/e/kubernetes-minikube
gilang31@LAPTOP-F9B95IVJ:/mnt/e/kubernetes-minikube$ kubectl apply -f deployment.yaml
deployment.apps/web-nginx created
gilang31@LAPTOP-F9B95IVJ:/mnt/e/kubernetes-minikube$ kubectl apply -f service.yaml
service/web-nginx-service created
```

```
gilang31@LAPTOP-F9B95IVJ:/mnt/e/kubernetes-minikube$ kubectl get all
NAME                                READY   STATUS    RESTARTS   AGE
pod/web-nginx-6798694bb6-7rmbx      1/1     Running   0          68s
pod/web-nginx-6798694bb6-87nd2      1/1     Running   0          68s
pod/web-nginx-6798694bb6-qxghz      1/1     Running   0          68s

NAME                        TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
service/kubernetes          ClusterIP   10.96.0.1       <none>        443/TCP        31m
service/web-nginx-service   NodePort    10.110.109.64   <none>        80:30512/TCP   52s

NAME                        READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/web-nginx   3/3     3            3           68s

NAME                                   DESIRED   CURRENT   READY   AGE
replicaset.apps/web-nginx-6798694bb6   3         3         3       68s
```

```
gilang31@LAPTOP-F9B95IVJ:/mnt/e/kubernetes-minikube$ kubectl get pods
NAME                        READY   STATUS    RESTARTS   AGE
web-nginx-6798694bb6-7rmbx  1/1     Running   0          82s
web-nginx-6798694bb6-87nd2  1/1     Running   0          82s
web-nginx-6798694bb6-qxghz  1/1     Running   0          82s
```

```
gilang31@LAPTOP-F9B95IVJ:/mnt/e/kubernetes-minikube$ kubectl get pods -w
NAME                          READY   STATUS    RESTARTS   AGE
web-nginx-6798694bb6-7rmbx    1/1     Running   0          3m22s
web-nginx-6798694bb6-87nd2    1/1     Running   0          3m22s
web-nginx-6798694bb6-qxghz    1/1     Running   0          3m22s
```

4. Akses Aplikasi via Browser

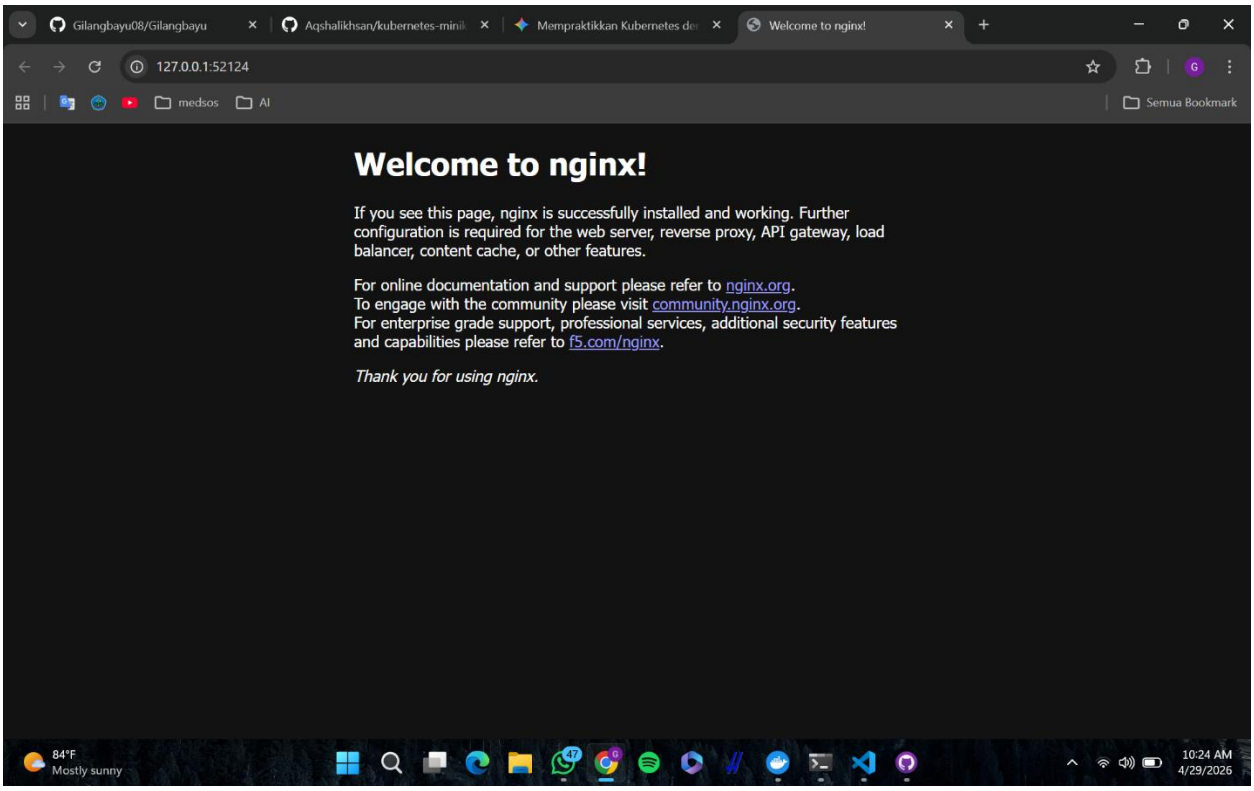
```
PS C:\Users\GILANG BAYU> minikube service web-nginx-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	web-nginx-service	80	http://192.168.49.2:30512

```
🔗  Starting tunnel for service web-nginx-service.
```

NAMESPACE	NAME	TARGET PORT	URL
default	web-nginx-service		http://127.0.0.1:57857

```
🎉  Opening service default/web-nginx-service in default browser...
❗  Because you are using a Docker driver on windows, the terminal needs to be open to run it.
```



5. Uji Skalabilitas (Scaling Pods)

```
PS E:\kubernetes-minikube> kubectl get pods -w
NAME                          READY   STATUS    RESTARTS   AGE
web-nginx-6798694bb6-7rmbx    1/1     Running   0          9m50s
web-nginx-6798694bb6-87nd2    1/1     Running   0          9m50s
web-nginx-6798694bb6-qxghz    1/1     Running   0          9m50s
web-nginx-6798694bb6-lqrvc    0/1     Pending   0          0s
web-nginx-6798694bb6-kq8ks    0/1     Pending   0          0s
web-nginx-6798694bb6-lqrvc    0/1     Pending   0          0s
web-nginx-6798694bb6-kq8ks    0/1     Pending   0          0s
web-nginx-6798694bb6-lqrvc    0/1     ContainerCreating   0
   0s
web-nginx-6798694bb6-kq8ks    0/1     ContainerCreating   0
   0s
web-nginx-6798694bb6-lqrvc    1/1     Running             0
   5s
web-nginx-6798694bb6-kq8ks    1/1     Running             0
   7s
```

```
PS E:\kubernetes-minikube> kubectl scale deployment web-nginx --r
eplicas=5
deployment.apps/web-nginx scaled
PS E:\kubernetes-minikube>
```

```
PS E:\kubernetes-minikube> kubectl scale deployment web-nginx --
eplicas=3
deployment.apps/web-nginx scaled
PS E:\kubernetes-minikube>
```

```
PS E:\kubernetes-minikube> kubectl get pods -w
NAME                        READY   STATUS    RESTARTS   AGE
web-nginx-6798694bb6-7rmbx  1/1     Running   0          9m50s
web-nginx-6798694bb6-87nd2  1/1     Running   0          9m50s
web-nginx-6798694bb6-qxghz  1/1     Running   0          9m50s
web-nginx-6798694bb6-lqrvc  0/1     Pending   0          0s
web-nginx-6798694bb6-kq8ks  0/1     Pending   0          0s
web-nginx-6798694bb6-lqrvc  0/1     Pending   0          0s
web-nginx-6798694bb6-kq8ks  0/1     Pending   0          0s
web-nginx-6798694bb6-lqrvc  0/1     ContainerCreating   0          0s
web-nginx-6798694bb6-kq8ks  0/1     ContainerCreating   0          0s
web-nginx-6798694bb6-lqrvc  1/1     Running             0          5s
web-nginx-6798694bb6-kq8ks  1/1     Running             0          7s
web-nginx-6798694bb6-kq8ks  1/1     Terminating         0          84s
web-nginx-6798694bb6-lqrvc  1/1     Terminating         0          84s
web-nginx-6798694bb6-kq8ks  1/1     Terminating         0          84s
web-nginx-6798694bb6-lqrvc  1/1     Terminating         0          118s
web-nginx-6798694bb6-lqrvc  0/1     Completed           0          85s
web-nginx-6798694bb6-kq8ks  0/1     Completed           0          85s
web-nginx-6798694bb6-kq8ks  0/1     Completed           0          86s
web-nginx-6798694bb6-kq8ks  0/1     Completed           0          86s
web-nginx-6798694bb6-kq8ks  0/1     Completed           0          86s
web-nginx-6798694bb6-lqrvc  0/1     Completed           0          86s
web-nginx-6798694bb6-lqrvc  0/1     Completed           0          86s
```

```
PS E:\kubernetes-minikube> kubectl get deployment -w
NAME        READY   UP-TO-DATE   AVAILABLE   AGE
web-nginx   3/3     3            3           12m
```

6. Manajemen Infrastruktur (Tambah Worker Node)

```
PS C:\Users\GILANG BAYU> minikube node add
😄  Adding node m02 to cluster minikube as [worker]
❗  Cluster was created without any CNI, adding a node to it might cause broken networking.
👍  Starting "minikube-m02" worker node in "minikube" cluster
🚜  Pulling base image v0.0.50 ...
🔥  Creating docker container (CPUs=2, Memory=2200MB) ...
🐳  Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
🔎  Verifying Kubernetes components...
🏄  Successfully added m02 to minikube!
PS C:\Users\GILANG BAYU>
```

```
^Cgilang31@LAPTOP-F9B95IVJ:/mnt/e/kubernetes-minikube$ kubectl get nodes
NAME           STATUS     ROLES           AGE    VERSION
minikube       Ready      control-plane   47m    v1.35.1
minikube-m02   NotReady   <none>          102s   v1.35.1
gilang31@LAPTOP-F9B95IVJ:/mnt/e/kubernetes-minikube$
```

```
PS C:\Users\GILANG BAYU> minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

minikube-m02
type: Worker
host: Running
kubelet: Running

PS C:\Users\GILANG BAYU>
```

7. Pembersihan Resource (Cleanup)
   Menghapus semua resource yang telah dibuat agar tidak membebani memori laptop.

```
gilang31@LAPTOP-F9B95IVJ:/mnt/e/kubernetes-minikube$ kubectl delete -f deployment.yaml
deployment.apps "web-nginx" deleted from default namespace
gilang31@LAPTOP-F9B95IVJ:/mnt/e/kubernetes-minikube$ kubectl delete -f service.yaml
service "web-nginx-service" deleted from default namespace
gilang31@LAPTOP-F9B95IVJ:/mnt/e/kubernetes-minikube$
```